

Class Adaptive Threshold Re-weighting Loss for Small Object Detection in Automotive SAR

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Abstract—With low atmospheric attenuation and better penetration of rain, smoke and dust, millimeter-wave radar perceptions have prevailed upon darkness and extreme weather conditions. As integrating SAR technology into data representation, image-based networks (such as YOLO and ResNet) are utilized to detect vehicles, motor-cycles and static objects. For object detection on automotive SAR images, conventional YOLO algorithms can effectively classify objects into distinct classes. However, specific to complex road conditions, the probability of false detections are increased with strong clutters and specular reflection, which make the SAR generate partial or weak image clips of the detected objects, especially in the small obstacles. We observe the under-learned and limiting detection accuracy for difficult classes is caused by attention imbalance during training. To address this issue, CATR Loss (Class-Adaptive Threshold Reweighting Loss), which adaptively defines sample difficulty based on predicted confidence, is proposed. By applying a class-wise dynamic threshold and incorporating a matched-aware quality constraint, the training attention can be shifted toward under-optimized positives. Experimental results show that CATR Loss integrated into YOLOv11s improves precision, recall, and mAP consistently without added inference overhead.

Index Terms—synthetic aperture radar, object detection, adaptive reweighting, class loss

I. INTRODUCTION

In order to create safer and more efficient roadways, one of the key factors is the advanced development of reliable sensor perception for autonomous driving systems. Under low-light and adverse weather conditions, radar has shown robust perception abilities and superior effectiveness due to its low cost and small size [1], [2]. With SAR technology, radar visions are extended into high resolution image representation, which informs not only the 2D position of the objects, but also their contours. Compared to LiDAR [3] and camera-based object detectors [4], scene understanding in SAR images depends highly on annotations with precise bounding boxes.

This work was supported in part by the National Natural Science Foundation of China under Grant 62571342, Grant 62431021, the Guangdong Basic and Applied Basic Research Foundation Grant. 2025B1515020076; the Foundation of Shenzhen City under Grant JCYJ20230808105359045; the Key Project of International Cooperation and Exchanges of the National Natural Science Foundation of China under Grant 62220106009, the project of Shenzhen Peacock Plan Teams under Grant KQTD20210811090051046, the Shenzhen University 2035 Program for Excellent Research under Grant 2022B009; Research Team Cultivation Program of Shenzhen University under Grant 2023DFT003; the Project of Department of Education of Guangdong Province (No. 2021KCXTD008), and the Shenzhen Key Laboratory of Advanced Navigation Technology under Grant ZDSYS201507081625213 .

Unlike traditional radar tensor, SAR-stylized representations have physical interpretations, thereby the learned features from image-based algorithms can be translated into SAR image understanding. The representative one-stage object detectors, YOLO series [5], are utilized to model the textural features and their correlations in SAR images, which gain high precision and recall performance. However, the clutter-sensitive and strong reflections from stationary objects make weak perception of small objects in SAR images [6], [7]. Additionally, complex road conditions will exacerbate the problem by generating partial or noisy objects in the presence of noise and interferences.

As for identifying small objects in SAR images, it is essential to realize an attention imbalance is occurred during the training process [8]. Motivated by YOLO models, objects are represented as 2D bounding boxes (x,y,h,w,c) on projection plane, where (x,y) is the bounding box center, h and w are height and width of the bounding box, while c is the class label [9]. Instead of leveraging one-hot embedding, YOLOv11 refines the position and predicts the classification of each object by probability scored from a matching principle.

For the efficiency and simplicity, YOLO-based detectors have been widely used to capture semantic information in SAR images [10]. However, positive samples with highly confidence, referred to as "easy positives", have gained extensive gradient update, whereas the "hard positives" with relative low predictions are distributed biased training attentions, which may cause limiting ability in learning to detect those small, difficult categories [11]–[13]. Accordingly, recent studies have explored both loss-form redesign and dynamic reweighting to alleviate training imbalance. By dynamically reweighting samples based on the intersection over union (IoU) [14], slide loss mitigate late-stage training imbalance. Leng proposes Poly Loss [15], a polynomially parameterized objective that generalizes cross-entropy and Focal loss [16], which explores a broader optimization. Li introduces an automated loss-search strategy guided by evaluation loss curves obtained by testing multiple loss functions, to reduce manual tuning effort [17]. Most efforts have been made to modify the global loss function or to employ sample-level re-weighting, however, the dominant dilemma lies inside the positive set, where easy positives dominate gradients and hard positives receive insufficient optimization, ultimately limiting learning for difficult categories and small targets.

In this paper, we propose a novel CATR Loss (Class-Adaptive Threshold Reweighting Loss) for the refinement in YOLO classification. Without introducing any additional inference overhead, CATR Loss adjusts training attention by imposing dynamic matching scores from class-specific confidence measurement. The model architecture and inference graph can be maintained, as well the small objects can be well learned. The thresholds and difficulty-amplification strengths are updated once per epoch. Extensive experiments have been conducted on an automotive SAR dataset, termed as AutoSAR [18].

Compared with the BCE baseline, CATR Loss brings consistent improvements on the test set while keeping the inference cost unchanged. Specifically, the overall Precision increases from 0.8255 to 0.8547 and the Recall increases from 0.7369 to 0.7876, thereby leading to a clear gain on mAP, i.e., mAP@0.5 rises from 0.7850 to 0.8249 and mAP@0.5:0.95 rises from 0.4773 to 0.5037. These improvements are mainly contributed by small targets and the difficult classes, which are usually the main source of missed detections in vehicular SAR scenes.

II. METHODOLOGY

In the classification branch of the YOLO detection head, the model outputs C classification logits per location. With N flattened locations, the logit for the i -th location and c -th class is $z_{i,c}$, with predicted probability $p_{i,c} = \sigma(z_{i,c}) = \frac{1}{1+e^{-z_{i,c}}}$ [19]. YOLO uses soft supervision $y_{i,c}$ from an alignment based matcher, that is the alignment score that this location is a positive sample of class c [20]. Training is limited by negative-positive count imbalance and intra-positive gradient imbalance: across classes and target scales, easy positives (with higher p) dominate mid-to-late gradients, suppressing low-confidence hard positives and slowing convergence while reducing recall for difficult and small-target classes. Focal-style modulation typically measures difficulty via $p_t = y \cdot p + (1 - y) \cdot (1 - p)$ [21], but under soft supervision y is small and unstable early and p_t can be incorrectly overestimated, even approaching 1, inducing pseudo-easy bias and down-weighting truly hard positives. We instead take $p_{i,c} = \sigma(z_{i,c})$ as the difficulty signal and propose CATR Loss, introducing an adaptive modulation factor only for positive terms: without changing the assigner, the positive/negative sets, or the normalization scheme, to reallocate the gradient budget within the positive set. The classification loss is written as

$$L_{cls} = \frac{\sum_{i,c} m_{i,c} L_{bce}(z_{i,c}, y_{i,c})}{\sum_{i,c} (y_{i,c} + \varepsilon)}, \quad (1)$$

$$m_{i,c} = \begin{cases} \omega'_{i,c}, & y_{i,c} > 0, \\ 1, & y_{i,c} = 0. \end{cases}$$

The parameter ε used to prevent division by zero is set to 1×10^{-12} . Negatives use fixed weight $m_{i,c} = 1$, while each positive weight $\omega'_{i,c}$ is determined by a class-wise dynamic threshold τ_c and a class-wise difficulty coefficient k_c , where

τ_c delineates the hard-sample regime and k_c controls amplification strength. Positive weights are computed using $\{\tau_c, k_c\}$ from the previous epoch; after each epoch, quality-weighted statistics of the p distribution update $\{\tau_c, k_c\}$. The following presents positive-weight construction and normalization, computation of τ_c and k_c .

A. Adaptive positive-sample weighting $\omega_{i,c}$

This section derives the explicit CATR Loss used to reweight element-wise classification BCE terms per batch. Adaptive modulation is applied only to positive terms, while negative terms keep weight 1, restricting gradient reallocation to the positive set and maintaining stable background suppression; positive weights are scale-normalized within positive terms to keep the overall calibration of the classification loss stable. Let the logit of the i -th location for the c -th class be $z_{i,c}$ with predicted probability $p_{i,c} = \sigma(z_{i,c})$, and let the alignment-based matcher output $y_{i,c}$ as supervision, i.e., the alignment score that this location is a positive of class c ; define the positive set as $\Omega = \{(i, c) | y_{i,c} > 0\}$. To eliminate early-stage bias under supervision based on p_t , difficulty is computed from $p_{i,c}$ relative to the class-adaptive threshold τ_c (Section B), avoiding erroneous down-weighting caused by pseudo-easy samples in early training. For any $(i, c) \in \Omega$, define continuous hardness as

$$h_{i,c} = \frac{\max(\tau_c - p_{i,c}, 0)}{\tau_c + \varepsilon}, \quad (2)$$

it is zero when $p_{i,c} \geq \tau_c$, and increases as $p_{i,c}$ decreases when $p_{i,c} < \tau_c$, indicating a hard sample. With class-adaptive difficulty coefficient $k_c \geq 0$ (Section C), define difficulty weight

$$\omega_{i,c}^{\text{hard}} = 1 + k_c h_{i,c}, \quad (i, c) \in \Omega, \quad (3)$$

so harder positives receive larger weights, with an upper bound determined by k_c . Matching quality is incorporated via quality weight

$$q_{i,c} = q_0 + (1 - q_0) \sqrt{y_{i,c} + \varepsilon}, \quad (i, c) \in \Omega, \quad (4)$$

with floor $q_0 \in [0, 1]$; a square-root transform of $y_{i,c}$ to avoid excessive influence from extremely small or extremely large $y_{i,c}$ values on the quality weight. Multiplying the difficulty and quality weights yields the raw positive weight

$$\omega_{i,c} = \omega_{i,c}^{\text{hard}} q_{i,c}, \quad (i, c) \in \Omega. \quad (5)$$

To prevent loss-scale drift as the hard-sample proportion varies across batches, mean-normalize positive weights over Ω as

$$\bar{\omega} = \frac{1}{|\Omega|} \sum_{(i,c) \in \Omega} \omega_{i,c}, \quad (6)$$

$$\omega'_{i,c} = \frac{\omega_{i,c}}{\bar{\omega} + \varepsilon}, \quad (i, c) \in \Omega, \quad (7)$$

define the final modulation factor as

$$m_{i,c} = \begin{cases} \omega'_{i,c}, & (i, c) \in \Omega, \\ 1, & y_{i,c} = 0, \end{cases} \quad (8)$$

and apply it to the element-wise BCE loss as

$$L'_{i,c} = m_{i,c} L_{\text{bce}}(y_{i,c}, z_{i,c}). \quad (9)$$

Without changing the loss form or the normalization scheme, CATR Loss amplifies hard positives while maintaining stability via quality constraints and within-set normalization.

B. Class-wise adaptive threshold τ_c

This section defines the class-wise adaptive threshold τ_c , which determines given a class's current learning progress, which low positives are treated as hard examples. During the e -th epoch, statistics are accumulated to compute a raw threshold τ_c^{raw} at epoch end; it is then constrained by a slow-update mechanism and applied in the next epoch. For each class c , we collect all positive terms $\Omega_c = \{i | y_{i,c} > 0\}$ and use $\sqrt{y_{i,c}}$ as weights to compute the class-wise weighted mean predicted probability

$$\mu_c^{(e)} = \frac{\sum_{i \in \Omega_c} \sqrt{y_{i,c}} p_{i,c}}{\sum_{i \in \Omega_c} (\sqrt{y_{i,c}} + \varepsilon)}. \quad (10)$$

To reduce per-epoch noise, we apply an exponential moving average $\bar{\mu}_c^{(e)} = \lambda \cdot \mu_c^{(e)} + (1 - \lambda) \cdot \bar{\mu}_c^{(e-1)}$ with smoothing coefficient $\lambda \in (0, 1)$ [22]. Given fixed global hyperparameter target mean level μ_{target} , we define the learning gap

$$\text{gap } p_c^{(e)} = \max\left(0, \frac{\mu_{\text{target}} - \bar{\mu}_c^{(e)}}{\mu_{\text{target}} + \varepsilon}\right), \quad (11)$$

which increases when $\bar{\mu}_c^{(e)}$ falls below the target, indicating an under-learned class. The hard-sample ratio is then set to

$$\rho_c^{(e)} = \rho_{\text{base}} + \rho_{\text{gain}} \text{gap } p_c^{(e)}, \quad (12)$$

where ρ_{base} is the base hard ratio and ρ_{gain} is the gap gain. With this ratio, we sort $\{p_{i,c} | i \in \Omega_c\}$ in ascending order to obtain $p_{(1),c} \leq p_{(2),c} \leq \dots \leq p_{(M),c}$ with weights $\sqrt{y_{(j),c}}$, compute cumulative weight

$$\begin{aligned} S_{k,c} &= \sum_{j=1}^k \sqrt{y_{(j),c}}, \\ S_{M,c} &= \sum_{j=1}^M \sqrt{y_{(j),c}}, \end{aligned} \quad (13)$$

and define the raw threshold τ_c^{raw} as the smallest predicted probability satisfying the weighted-quantile condition [23]

$$\begin{aligned} k^* &= \min \left\{ k \mid S_{k,c} \geq \rho_c^{(e)} S_{M,c} \right\}, \\ \tau_c^{\text{raw}} &= p_{(k^*),c}. \end{aligned} \quad (14)$$

Thus, $\rho_c^{(e)}$ controls the fraction of positives deemed hard and τ_c adapts to class-specific learning progress: when a class is under-learned (lower $\bar{\mu}_c$), ρ_c increases, raising τ_c and expanding the hard-sample proportion; when learning is sufficient, ρ_c remains low, limiting disturbance to easy positives.

C. Class-wise adaptive difficulty coefficient k_c

The class-wise adaptive difficulty coefficient k_c controls hard-positive amplification and enters the difficulty weight linearly as $\omega_{i,c}^{\text{hard}} = 1 + k_c \cdot h_{i,c}$, $(i, c) \in \Omega$. To strengthen under-learned classes while avoiding overly aggressive amplification under unreliable statistics (e.g., scarce samples or high noise), we define k_c as a joint function of the learning gap and sample reliability, update it in the current epoch, and apply it in the next epoch so it is fixed within each epoch and varies smoothly across epochs. We assign statistical weights $\sqrt{y_{i,c}} + \varepsilon$, $i \in \Omega_c$ to class positives, define effective statistical size $W_c^{(e)} = \sum_{i \in \Omega_c} \sqrt{y_{i,c}} + \varepsilon$, and construct reliability coefficient

$$\text{rel}_c^{(e)} = \sqrt{\frac{W_c^{(e)}}{W_c^{(e)} + \omega_{\text{ref}}}}, \quad (15)$$

with reference scale ω_{ref} . When the effective positive-sample weight is small ($W_c^{(e)} \ll \omega_{\text{ref}}$), $\text{rel}_c^{(e)} \approx 0$, indicating unreliable statistics and requiring strong suppression of amplification; when statistics are sufficient ($W_c^{(e)} \gg \omega_{\text{ref}}$), $\text{rel}_c^{(e)} \rightarrow 1$, allowing amplification to follow the learning gap. The coefficient is then defined as

$$k_c^{(e)} = k_{\text{base}} + k_{\text{gain}} \text{gap } p_c^{(e)} \text{rel}_c^{(e)}, \quad (16)$$

where k_{base} is the base amplification strength and k_{gain} is the gap gain. As $\text{gap } p_c^{(e)}$ increases, $k_c^{(e)}$ increases, strengthening hard-sample weights in the next epoch; when $\text{rel}_c^{(e)}$ is small, even large $\text{gap } p_c^{(e)}$ yields a strongly suppressed $k_c^{(e)}$, preventing excessive weights driven by unreliable statistics.

III. EXPERIMENTS

A. Experimental setup

We validate the proposed method on a published vehicular SAR target detection dataset [18]. All the experimental settings are same except that the original BCE in the classification loss of YOLOv11s is replaced with CATR Loss. The input size is 640×640 ; training runs for 500 epochs with batch size 64, using SGD with initial learning rate 0.01, and data augmentation follows the default YOLOv11s configuration. The smoothing parameter λ is set to 0.1, and the lower bound of the quality weight q_0 is set to 0.2. All experiments are conducted on an NVIDIA RTX A6000.

B. Experimental results

In Fig. 1, after using CATR Loss, Precision rising up for almost every class except handrail. The biggest increases are on small targets: person rises from 0.4620 to 0.5978, and light rises from 0.7035 to 0.7551. Recall shows the same pattern: person increases the most, which increases from 0.4713 to 0.6707. A few classes that were already doing well drop slightly in Recall: car (-0.0081), building (-0.0435), and tree (-0.0379), but their Precision stays high. Overall, the changes mostly raise Recall, while Precision is kept the same or higher for most classes; only a few strong classes show a small drop in Precision.

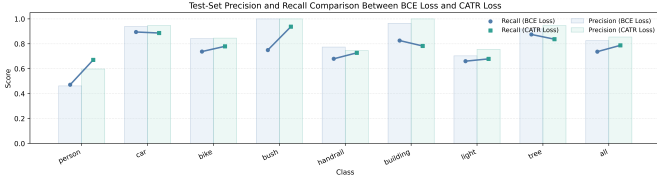


Fig. 1: Comparison of Precision and Recall on the test set between BCE Loss and CATR Loss. The bar chart shows Precision, and the line plot shows Recall.

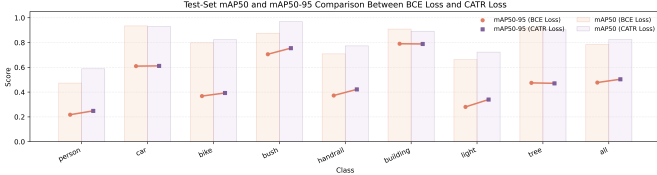
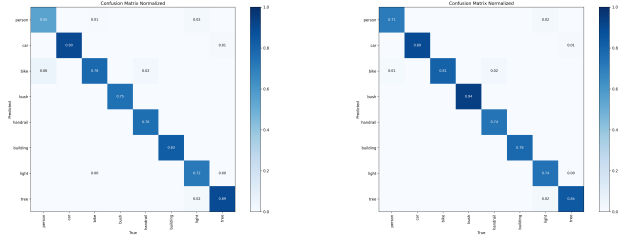


Fig. 2: Comparison of mAP50 and mAP50-95 on the test set between BCE Loss and CATR Loss. The bar chart shows mAP50, and the line plot shows mAP50-95.

Fig. 2 shows that with CATR Loss, the mean mAP@0.5 increases from 0.7850 to 0.8249, and the mean mAP@0.5:0.95 increases from 0.4773 to 0.5037. Most of the gain comes from the harder classes; for example, person improves from 0.4730 to 0.5882 in mAP@0.5 and from 0.2177 to 0.2485 in mAP@0.5:0.95. A few already strong classes show small drops in mAP@0.5 (e.g., car, building, and tree), but the changes are minor and do not change the overall picture. In short, the update mainly helps difficult classes and tougher scenes.



(a) Normalized confusion matrix of BCE Loss. (b) Normalized confusion matrix of CATR Loss.

Fig. 3: Normalized confusion matrix comparison.

Fig. 3 shows the normalized confusion matrix after replacing BCE Loss with CATR Loss. The biggest improvement is on the harder classes: person rises from 0.55 to 0.71 and bush from 0.75 to 0.94. Bike and light also go up, from 0.78 to 0.81 and from 0.72 to 0.74. A few classes that were already strong drop a bit, including car (-0.01), handrail (-0.02), and tree (-0.05); building drops more, from 0.83 to 0.78 (-0.05). Person incorrectly detected as bike drops from 0.06 to 0.01. Some confusions are still there, like light and tree (about 0.02) and handrail incorrectly detected as bike (about 0.02). This behavior is expected because CATR redistributes the gradient budget toward hard positives and under-learned classes. As

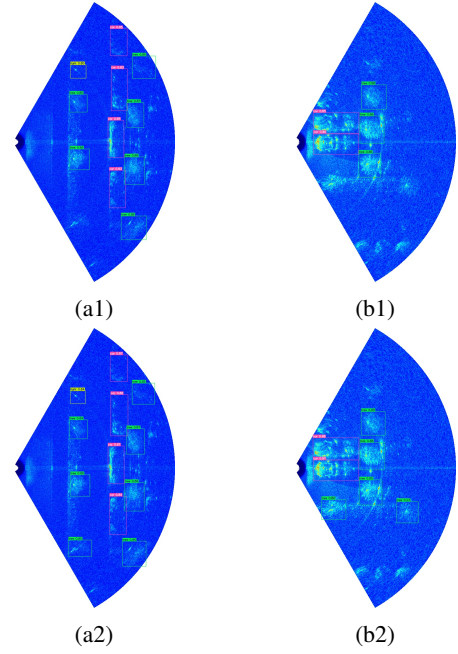


Fig. 4: Detection results on two SAR scenes. For each scene, the top image is BCE Loss and the bottom image is CATR Loss.

a result, a few already well-optimized dominant classes may show slight fluctuations.

Fig. 4 shows two clear changes after replacing BCE Loss with CATR Loss: (i) higher confidence on true targets, and (ii) fewer missed detections. Across the examples, true-positive confidence goes up. For example, car increases by about 0.01–0.02, light by 0.04 in scene (a). We also recover missed objects in scenes (a) and (b) with useful confidence like tree(0.80 and 0.23). Put together, Fig. 4 shows stronger true detections and fewer low-confidence clutter boxes, which fits the goal of putting more weight on hard positives.

IV. CONCLUSION

We study a training imbalance problem in vehicular synthetic aperture radar (SAR) object detection: easy positives dominate the classification gradients, and hard positives get too little learning. To tackle this, we propose CATR Loss and test it in YOLOv11s. CATR measures how hard a positive is using the classification score and a per-class moving threshold, then reweights only the positive classification terms; a match-quality constraint limits the impact of low-quality matches. It does not change inference. On the vehicular SAR dataset, CATR improves performance, with most of the gain coming from difficult classes and cluttered or small-target scenes, while the overall confusion pattern stays about the same. Currently, the values of the hyperparameters are primarily determined through grid search experiments. Next, we will further conduct an exploration of the parameter space and perform a sensitivity analysis on the hyperparameters. We will compare against more loss designs, test across datasets and conditions, and take a closer look at how stable the per-class updates are and how well they carry over.

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